Math 514, F13 **Name:** Answer

Quiz 12 **SSN:** xxx-xx-

Instructions: You must show all of your work to receive a credit for a correct answer.

Problem 1. (10 points) The prices of a security follow a geometric Brownian motion with drift parameter $\mu = 0.05$ and volatility parameter $\sigma = 0.4$. The security's present price is 200. A new investment that is being marketed costs 20; after 2 year the investment will pay 25 if $S(2) < 180$, will pay x if $S(2) > 210$, and will pay 0 otherwise. The nominal interest rate is 6%, continuously compounded.

(a) What must be the value of x if this new investment is not to give rise to an arbitrage?

$(r-\sigma^2/2)t = (0.06-0.08)\times 2 = -0.04$, $\sigma\sqrt{t} = 0.4\times\sqrt{2} = 0.5656$

Let W be the normal random variable with mean -0.04 and standard deviation 0.5656. We then have.

$$20 = \left(25P(S(2)<180) + xP(S(2)>210)\right)e^{-0.06\times 2}$$

$$e^{0.12}20 = 25P\left(W<\log\left(\tfrac{180}{200}\right)\right) + xP\left(W>\log\left(\tfrac{210}{200}\right)\right)$$

$$e^{0.12}20 = 25P\left(Z<\frac{\log(0.9)+0.04}{0.5656}\right) + xP\left(Z>\frac{\log(1.05)+0.04}{0.5656}\right)$$

$$e^{0.12}20 = 25\Phi(-0.1157) + x(1-\Phi(0.1572))$$

$$x = \frac{20e^{0.12} - 25\Phi(-0.1157)}{1-\Phi(0.1572)} \approx 25.6$$

(b) What is the probability that $S(1) > 180$?

$\mu t = 0.05\times 1 = 0.05$. $\sigma\sqrt{t} = 0.4\times 1 = 0.4$.

Let W be the normal random variable with mean 0.05 and standard deviation 0.4.

$$P\{S(1)>180\} = P\left(W>\log\left(\tfrac{180}{200}\right)\right)$$

$$= P\left(Z>\frac{\log(0.9)-0.05}{0.4}\right)$$

$$= 1-\Phi(-0.2304)$$

$$\approx 0.5911$$